

Heterogeneous Interactions between Gas-Phase Pyruvic Acid and Hydroxylated Silica Surfaces: A Combined Experimental and Theoretical Study

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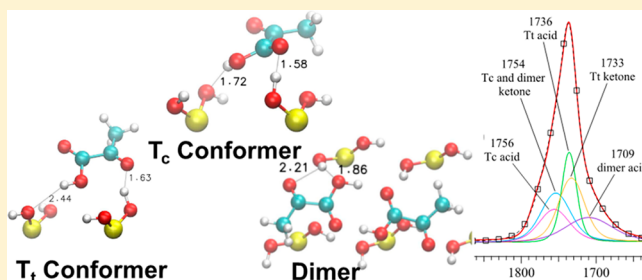
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ABSTRACT: The adsorption of gas-phase pyruvic acid ($\text{CH}_3\text{COCO}_2\text{H}$) on hydroxylated silica particles has been investigated at 296 K using transmission Fourier transform infrared (FTIR) spectroscopy and theoretical simulations. Under dry conditions (<1% relative humidity, RH), both the trans–cis (Tc) and trans–trans (Tt) pyruvic acid conformers are observed on the surface as well as the (hydrogen bonded) pyruvic acid dimer. The detailed surface interactions were further understood through ab initio molecular dynamics simulations. Under higher relative humidity conditions (above 10% RH), adsorbed water competes for surface adsorption sites. Adsorbed water is also observed to change the relative populations of the different adsorbed pyruvic acid configurations. Overall, this study provides valuable insights into the interaction of pyruvic acid with hydroxylated silica surfaces on the molecular level from both experimental and theoretical analyses. Furthermore, these results highlight the importance of the environment (relative humidity and coadsorbed water) in the adsorption, partitioning, and configurations of pyruvic acid at the surface.



INTRODUCTION

Pyruvic acid, $\text{CH}_3\text{COCO}_2\text{H}$, an important atmospheric α -keto, is formed in the atmosphere primarily by direct photolysis and secondarily by particle-phase aqueous reaction between hydroxyl radical and hydrated methylglyoxal (an abundant oxidation product of precursors such as isoprene and aromatic compounds).¹ Numerous field studies have detected pyruvic acid in the gas phase (10–100 ppt), in aerosols (up to 140 ng·m⁻³), in snow, and in rainwater.^{2–5,5–12} A variety of reactions have been investigated for pyruvic acid including gas- and condensed-phase photolysis,^{13–22} hydroxyl radical oxidation,^{23–25} thermal decomposition,^{26,27} and vibrational overtone induced decarboxylation.^{28–30} Computational studies^{31–33} on pyruvic acid's electronic structure, conformation possibilities, and barriers to decarboxylation provide insights into experimental findings.

The structure of pyruvic acid is interesting as it is an α -dicarbonyl acid that has several conformers due to the intramolecular rotation around the C–C and C–O bonds.²⁸

Reva et al.³⁴ investigated the infrared spectroscopic features of the rotational conformers of pyruvic acid. Additionally, these conformers have also been studied using theoretical approaches,^{26,33–37} and for gaseous pyruvic acid, the two lowest energy conformers are separated by only 2.08 kcal·mol⁻¹, while other conformers lie at much higher energy.^{26,33,34,36–38} The lowest energy conformer is labeled as the trans–cis (Tc) conformer, while the higher energy conformer is the trans–trans (Tt) conformer. The Tt conformer has the acidic hydrogen rotated away from the ketonic oxygen while the most stable conformer form, the Tc conformer, forms an intramolecular hydrogen bond between the acidic hydrogen and the α -carbonyl. Since the Tt conformer is close enough in energy to the most stable Tc conformer, there is a significant thermal population of both at

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ambient temperatures.^{26,28,33,39–42} Plath et al.²⁸ revealed differences between the Tt and Tc conformers by studying the OH vibrational overtone transitions, and Reva et al.⁴² showed conformational switching between pyruvic acid Tt and Tc conformers by selective pumping of the first OH stretching vibration overtone.

Since mineral aerosol surfaces are ubiquitous in the atmosphere and play an important role in atmospheric chemistry,^{43–45} we are interested in how organic acids such as pyruvic acid interacts with these surfaces. Silica (SiO₂), an oxide with abundant surface hydroxyl groups, is an important component of mineral dust particles in the atmosphere.⁴⁶ Silica can also represent glass surfaces present in an indoor environment as pyruvic acid has been identified in human emanation and recently detected as one of the dominant gas-phase carboxylic acids in occupied classrooms.⁴⁷ Earlier studies have investigated the heterogeneous reaction of organic acids with silica.^{46,48–53} The uptake of organic acids including HCOOH and CH₃COOH on SiO₂ has been examined previously,^{46,52,53} as well as inorganic acids such as HNO₃.⁴⁹ Since there is an abundance of water vapor in the atmosphere, the role of coadsorbed water in the interaction of these acids with silica has been probed as well. These studies reveal that adsorbed water both competes for surface adsorption sites with HNO₃ and HCOOH and promotes their dissociation to hydronium ions and the corresponding anions.^{49,54} In the case of pyruvic acid, few studies have investigated the surface chemistry of this molecule; however, Schnitzler et al.⁵⁵ studied the intermolecular interactions between pyruvic acid with water complexes and suggested hydrogen bonded water affects pyruvic acid conformation and stability. In solution, it has been shown that the α -keto acid exists in the keto form but also with its geminal diol form;⁵⁶ the latter was not considered in this study, but the hydration could occur at high relative humidity.

The presence of water and electrolytes at the silica interface has been shown to drastically change the nature of the silanols acidity.^{57–60} The interaction of the pyruvic acid with silica and the role of water can be investigated using atomistic ab initio molecular dynamics simulations (AIMD). Simulations can provide the microscopic picture at the atomistic resolution and permit one to get insight into the experimental transmission Fourier transform infrared (FTIR) spectra and also can include the effect of the temperature, which was omitted in previous DFT calculations.^{28,34,38} A further advantage of ab initio simulations is that the heterogeneous environment of the pyruvic acid, including the solid (mineral) and solvent (liquid water), is included and treated consistently at the same level of theory.

Despite their importance for heterogeneous reactions, little is known about the interactions of pyruvic acid with surfaces, such as SiO₂. Our limited knowledge calls for a bottom-up approach where, starting from a dry condition, the bonding of pyruvic acid to the silica surface is investigated and the role of water solvent molecules is addressed progressively by increasing the level of humidity.

METHODS

Experimental Methods. The adsorption of pyruvic acid on silica surfaces as a function of pyruvic acid pressure at 296 $\pm$ 1 K and different relative humidity (RH) was studied using a modified Teflon coated infrared cell coupled with transmission Fourier transform infrared (FTIR) spectroscopy, as described in detail in previous studies.^{49,50,54} In these experiments,

approximately 5 mg of silica (Degussa), having a BET area of 230 m² g^{−1}, was pressed onto one-half of a tungsten grid held by two Teflon coated jaws in the sample cell compartment. The sample cell was then evacuated for 6 h using a turbo molecular pump to clean the cell and sample surface. After evacuation, the sample was exposed to the desired pressure of dry, gaseous pyruvic acid for 20 min under dry conditions (RH < 1%). Pyruvic acid vapor was taken from CH₃COCOOH (97%, Alfa Aesar) that had been first distilled and then degassed at least three times with consecutive freeze–pump–thaw cycles.

To investigate the effects of relative humidity, the oxide sample was first exposed to the desired pressure of pyruvic acid for 20 min, followed by the introduction of the desired pressure of water vapor (produced from degassed HPLC grade water, Fisher Chemicals) for an additional 20 min. Fixing the relative humidity while varying the pressure of pyruvic acid which the sample was exposed to allowed for the effects of pyruvic acid pressure to be studied. After each experiment, the IR cell was evacuated overnight using a turbo-molecular pump. Prior to and after the exposure of pyruvic acid, single beam spectra of the surface and gas phase (250 scans) were acquired at 296 K using a resolution of 4 cm^{−1} and covering the spectral range extending from 800 to 4000 cm^{−1}. As silica is opaque below 1200 cm^{−1}, spectra are shown only above 1200 cm^{−1}. For kinetic measurements, during and following exposure to pyruvic acid, single beam spectra of the silica surface (10 scans) were acquired using a Macro (OMNIC Macro Basics) for 60 s intervals. Absorbance spectra of pyruvic acid on silica particles are reported as the difference in the silica spectra before and after exposure to pyruvic acid. Absorption bands due to gas-phase pyruvic acid, measured through the blank half of the tungsten grid, were subtracted to obtain spectra of adsorbed pyruvic acid.

Numerical Methods. Born–Oppenheimer molecular dynamics simulations (BOMD) are carried out with the CP2K/Quickstep package.⁶¹ As a model for the silica surface, we used the fully hydroxylated (0001) α -quartz interface.^{57,58} The DFT (density functional theory) electronic representation is BLYP with Grimme D3 dispersion correction.⁶² GoedeckerTeterHutter (GTH) pseudopotentials⁶³ are used in conjunction with a plane-wave basis set defined by a density cutoff of 300 Ry and Gaussian basis sets of triple- ζ polarized type for all the atoms for which the TZV2P-MOLOPT functions are chosen. Orthorhombic boxes of dimensions 19.64 $\times$ 17.008 $\times$ 80 Å³ are used, with periodic boundary conditions applied in all three directions of space. Each simulated system is composed of 6 O–Si–O layers and 16 surface SiOH silanols of the Q² type (geminals). A gap of about 69 Å is left between periodic images of the slab in the z-direction. The top surface was used to examine the interaction with the pyruvic acid. We considered both the Tc and Tt pyruvic acid conformers (see Figure 1) to be initially placed parallel to the interface and at a vertical distance of 3 Å above the planar surface defined by the top oxygen atoms from the surface silanol group.

The following systems were investigated: (1) pyruvic acid alone; (2) pyruvic acid on dry silica (no physisorbed water molecules, 0% coverage) and; (3) pyruvic acid on the humid silica, namely, covered with one full monolayer of water (16 water molecules, 100% coverage).

BOMD were conducted in the canonical NVT ensemble. An equilibration time of 5 ps per trajectory for each interface was considered, followed by a production time for the analysis of

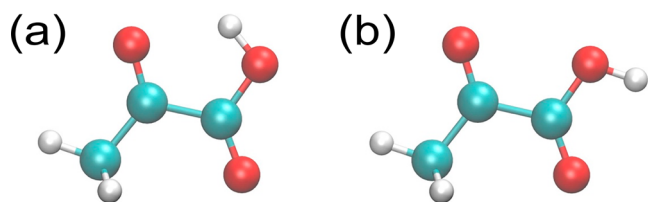


Figure 1. Optimized structures of Tc (a) and Tt (b).

15 ps with a time step 0.5 fs. For systems (1) and (2), the temperature of the thermostat is set at 298 K while for system (3) a slightly elevated temperature of 330 K is used in order to avoid the glassy behavior of BLYP liquid water. In these calculations, the silanol groups are free to reorient in response to the presence of the adsorbed molecules, namely, both pyruvic acid and water molecules (when present, due to the increased level of humidity). Indeed, the silanols do reorient in order, e.g., to better “solvate” the deprotonated pyruvic acid. Spontaneous deprotonation of silanol groups does not occur during the sampled trajectories.

We have also investigated the adsorption of dimers on the dry quartz interface. Two initial configurations were prepared by combining the previous optimized monomer: Tt–Tt and Tt–Ct with the internal hydrogen bonds between the carboxyl groups set initially at 1.67 Å⁶⁴ and with the dihedral angle $C_{\text{methyl}}-C_{\text{keto}}-C_{\text{acid}}-O_{\text{hydroxyl}}$ set to 0° for the cis form. The simulations were produced using the same previous protocol as for the monomers.

In order to calculate the binding energy, we performed the following minimization calculations using the BFGS minimizer. Along the equilibrium trajectory, we extracted configurations every 250 steps. For the monomer, two sets of calculation were performed: one with the monomer adsorbed on the surface and the other one where the monomer is away from the surface, at a height of 30 Å above the surface (gray inset in Figure 2). For the dimer, four calculations were performed: the

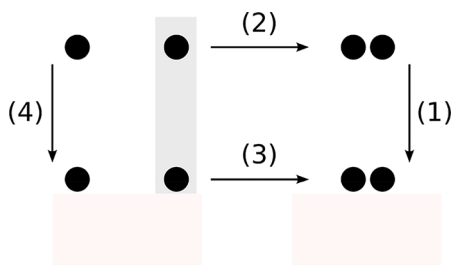


Figure 2. Diagram of the energy calculations: dimer binding energy to the surface (1), binding energy for the monomer (4), and dissociation energy of the dimer in vacuum (2) and on the surface (3).

first with the dimer adsorbed on the surface, the second with two monomers on the surface, separated by a half-cell distance, the third with the dimer in the gas phase displaced by $z = 30$ Å from the surface, and the fourth with two monomers displaced by $z = 30$ Å from the surface (see Figure 2).

RESULTS AND DISCUSSION

Adsorption of Pyruvic Acid on SiO₂ Surfaces. Dry Conditions: Experiment and Theory. It has been shown that the monomeric pyruvic acid exists under experimental temperature in gas phase with two geometries: the Tc conformer, having the lowest energy, and Tt, separated by

~2 kcal/mol from the latter.^{38,64–66} The most stable conformer (Tc) forms an intramolecular hydrogen bond between the acid hydrogen and the α -carbonyl which affects the vibrational spectrum by lowering the OH stretch frequency about ~120 cm^{−1} compared to the Tt conformer. The C=O stretching modes present also structural differences leading to a shifted frequency for the C=O stretch (see Figure 3).

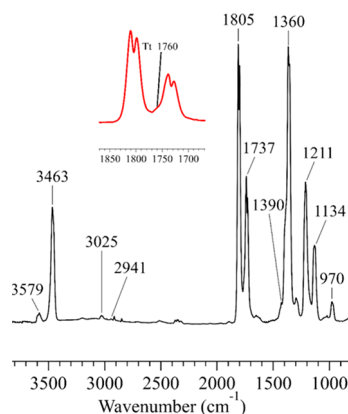


Figure 3. FTIR spectra of 1 Torr of gas-phase C₃H₄O₃(g) at 296 K in the spectral ranges extending from ca. 900 to 3800 cm^{−1} and inset from ca. 1660 to 1870 cm^{−1}. The small shoulder observed in the inset at 1760 cm^{−1} (shown in red) corresponds to the non-hydrogen bonded Tt conformer, while the main peak is due to the more stable intramolecular hydrogen bonded Tc conformer.

The FTIR spectrum of gas-phase pyruvic acid (C₃H₄O₃(g)) seen in Figure 3 was collected at 296 K for a pressure of 1 Torr (4 cm^{−1} resolution). The spectrum and the vibrational modes of gas-phase pyruvic acid are well understood.²⁸ The modes are assigned as follows: 3579 cm^{−1} for O–H stretching motion of the Tt conformer; 3463 cm^{−1} for O–H stretching motion of the Tc conformer; 1805 cm^{−1} for the C=O acid stretch; 1737 cm^{−1} for the C=O ketone stretch; 1360 cm^{−1} for the CH₃ symmetric bend; 1211 cm^{−1} for the COH bend; 1134 cm^{−1} for C–O stretch; 970 cm^{−1} for the CH₃ rock. Adjacent to the C=O ketone absorption (centered at 1737 cm^{−1}), a small shoulder (at 1760 cm^{−1}) is observed (see inset in Figure 3), corresponding to the non-hydrogen bonded Tt conformer. The main peak, however, is due to the more stable intramolecular hydrogen bonded Tc conformer.²⁸

The infrared spectra for hydroxylated silica particles after exposure to gas-phase pyruvic acid under dry condition (less than 1% RH) at 296 K and as a function of pyruvic acid pressure are displayed in Figure 4. The clean silica surface serves as the background reference spectrum. Observed infrared absorption band frequencies are assigned as follows: C=O stretch at 1737 cm^{−1}; CH₃ asymmetric bend at 1428 cm^{−1}; C–C asymmetric stretch at 1389 cm^{−1}; CH₃ symmetric bend at 1280 cm^{−1}. Pyruvic acid adsorption was also carried out at lower pyruvic acid pressures (5, 10, 15, and 25 mTorr) as shown in Figure 4b. A small shoulder (1776 cm^{−1}) to the left of the peak centered at 1737 cm^{−1} is clear at lower pyruvic acid coverages and can be assigned to the Tc conformer acid stretch, shifted from the gas-phase position (1805 cm^{−1}). The peak shift is due to weaker intramolecular hydrogen bonding following adsorption. Note that, at higher pressures, the peak at 1776 cm^{−1} merges into the highest intensity peak (centered at 1737 cm^{−1}) and becomes less distinct. Except for this

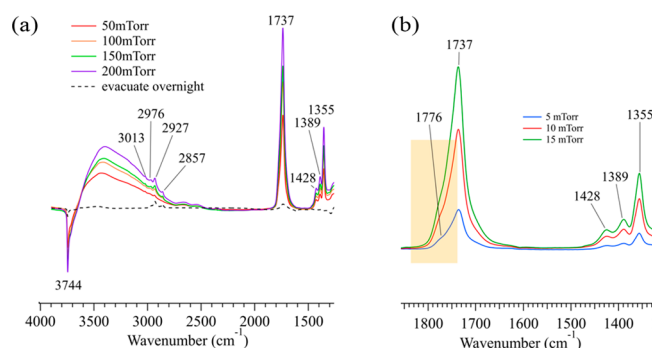


Figure 4. (a) Absorbance spectra of pyruvic acid adsorbed on silica under dry conditions (<1% RH) as a function of pressure in the spectral range extending from ca. 1280 to 4000 cm^{-1} . Note that SiO_2 is opaque below 1280 cm^{-1} due to lattice vibrations. Spectral subtraction was performed using gas-phase absorption spectra. Shown as a dashed line (— —) is the surface spectrum following overnight evacuation. (b) Absorbance spectra for lower pressures (5, 10, 15, and 25 mTorr) of pyruvic acid adsorbed on silica under dry conditions in the spectral range extending from ca. 1280 to 1850 cm^{-1} .

absorption band, all other observed band frequencies are in excellent agreement with that observed for gas-phase pyruvic acid and there is only a small wavenumber difference observed between the adsorbed and gas-phase pyruvic acid molecules. The negative peak at 3742 cm^{-1} is due to silica surface isolated silanol groups hydrogen bonded with pyruvic acid, with the broad band below 3600 cm^{-1} assigned to the hydrogen bonded Si—OH groups. Absorption peaks due to adsorbed pyruvic acid (1737, 1428, 1389, and 1355 cm^{-1}) disappear upon evacuation, while the negative peak attributed to isolated O—H groups (3742 cm^{-1}) reappears, suggesting that pyruvic acid is molecularly adsorbed on the silica surface via a reversible process. There is no spectral evidence for deprotonation of pyruvic acid to form pyruvate or for coupling reactions to form oligomers.

Ab initio molecular dynamics simulations of Tc and Tt conformers on the hydroxylated (0001) α -quartz interface reveal a molecularly adsorbed pyruvic acid through hydrogen bonding. In particular, we obtain a favorable adsorption energy of $-16.5 \text{ kcal}\cdot\text{mol}^{-1}$ for Tc and $-14.8 \text{ kcal}\cdot\text{mol}^{-1}$ for Tt. This finding suggests that the most stable conformer is also the one that more favorably binds to the surface. At finite temperature, due to the surface flexibility, several adsorption configurations are possible. Therefore, the calculated adsorption energies are the results of an average over ~ 100 snapshots extracted from the trajectory at room temperature.

The interactions of pyruvic acid with the surface, which we observe in the simulations, can help to rationalize the measured FTIR spectra. We find that the Tc conformer typically forms two hydrogen bonds (see Figure 5): one between the oxygen of the acidic carbonyl group and the hydrogen of a silanol group and the other one between the hydrogen of the carboxyl group and the oxygen of a silanol group. The methyl group oscillates between vacuum and the hydrophobic patches on the surface due to the thermal fluctuations. When forming an HB with a silanol group at the interface, the C=O bond is weakened. This results in a shift toward lower wavenumbers of the corresponding peak in the vibrational spectrum of about $\sim 38 \text{ cm}^{-1}$ compared to the vacuum value. On the other hand, since the C=O keto group does not interact with the interface, the corresponding peak in

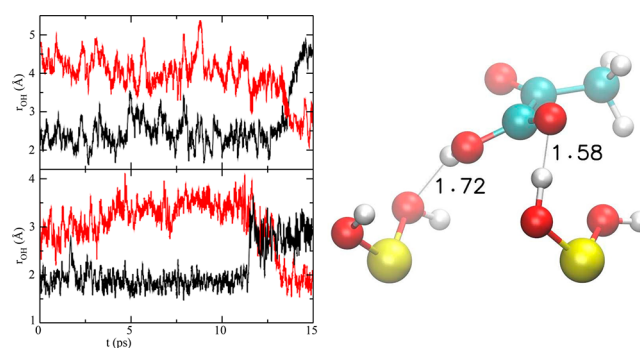


Figure 5. Hydrogen bond distance between the oxygen of the acidic carbonyl group and the hydrogen of the silanol groups (bottom panel) and between the hydrogen of the carboxyl group and the oxygen of the silanol groups (top panel) as a function of time. The black and red traces correspond to the different silanol groups shown on the right side. Right side represents a snapshot of the adsorbed Tc pyruvic acid.

the vibrational spectrum does not show any significant change in the frequency position with respect to the vacuum value (see Table 1).

Table 1. Vibrational Mode Assignment for Pyruvic Acid, CH_3COCOOH , Gas Phase and Adsorbed on Hydroxylated SiO_2

mode assignment	gas phase (cm^{-1})	SiO_2 (cm^{-1})	literature ⁶⁶ (cm^{-1})
$\nu(\text{OH, surface})$		3744	3742 ⁴⁹
$\nu(\text{OH, Tt})$	3579		3579
$\nu(\text{OH, Tc})$	3463		3463
$\nu_{\text{as}}(\text{CH}_3)$	3025, 2941	3013, 2976, 2927, 2857	3025, 2941
$\nu(\text{C=O}_{\text{acid}}, \text{Tc})$	1805	1776	1804
$\nu(\text{C=O}_{\text{acid}}, \text{Tt})$	1760	1756	1765, 1751 ⁴²
$\nu(\text{C=O}_{\text{keto}})$	1737	1737	1737
$\nu(\text{C=O}_{\text{dimer}})$		1710	1720, 1747 ⁶⁴
$\delta_{\text{as}}(\text{CH}_3)$	1422	1428	1424
$\nu_{\text{as}}(\text{C—C})$	1390	1389	1391
$\delta_{\text{s}}(\text{CH}_3)$	1360	1355	1360
$\delta(\text{COH})$	1211		1211
$\mu_{\text{as}}(\text{C—O})$	1134		1133

Similarly, the Tt conformer also forms two hydrogen bonds with the surface (see Figure 6). One is established between the hydrogen atom of the acidic carboxyl group and the oxygen from a silanol group, and the other one is formed between the oxygen of the C=O keto group and the hydrogen of a silanol group. In this case, the C=O acid group does not hydrogen bond and the corresponding stretching peak has the same frequency as obtained for the gas phase. The C=O keto group, instead, is involved in an hydrogen bond, producing a red shift of $\sim 29 \text{ cm}^{-1}$ in the corresponding stretching peak. As a result of the different interactions with the surface, the C=O keto group for Tc or Tt conformer presents a similar frequency value (see Table 2).

The simulation setup with one monomer on the surface gives a coverage of $3 \times 10^{13} \text{ molecules}/\text{cm}^2$ corresponding to the experiment performed at a pressure of 15 mTorr. At this low pressure, a shoulder is clearly visible in the experimental spectrum around 1776 cm^{-1} , shifted from the C=O acid stretching peak position of 29 cm^{-1} . According to the

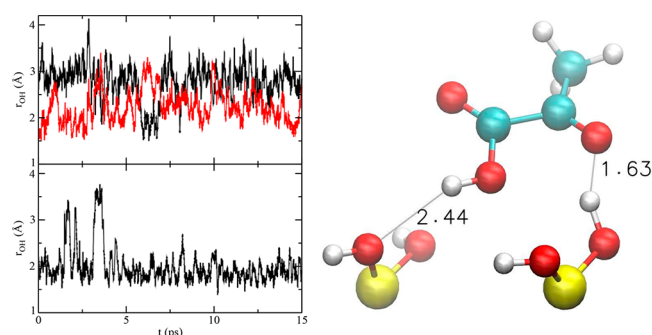


Figure 6. Hydrogen bond distance between the oxygen of the C=O keto group and the hydrogen of the silanol groups (bottom panel) and between the hydrogen of the carboxyl group and the oxygen of the silanol groups (top panel) as a function of time. The black and red traces correspond to different silanol groups shown on the right side. Right side represents a snapshot of the adsorbed Tt pyruvic acid.

Table 2. Calculated Vibrational Frequencies of Different Adsorbed Pyruvic Acid Species

species	mode	exp	gas	on dry surface	on wet surface
Tc	$\nu(\text{C}=\text{O}_{\text{acid}})$	1805	1734	1696	1651
	$\nu(\text{C}=\text{O}_{\text{ketone}})$	1737	1677	1676	1657
Tt	$\nu(\text{C}=\text{O}_{\text{acid}})$	1760	1697	1694	1629
	$\nu(\text{C}=\text{O}_{\text{ketone}})$		1702	1673	1668
Tt–Tt	$\nu(\text{C}=\text{O}_{\text{acid}})$			1610	
	$\nu(\text{C}=\text{O}_{\text{ketone}})$			1691	
Tt–Ct	$\nu(\text{C}=\text{O}_{\text{acid}})$			1633	
	$\nu(\text{C}=\text{O}_{\text{ketone}})$			1687	

simulations, such a shift can be assigned to the C=O keto stretch (the calculated shift from the simulations is $\sim 20 \text{ cm}^{-1}$).

As the pyruvic acid concentration increases, a question arises if pyruvic acid adsorption is also possible as a dimer. The dimerization in gas phase is strongly favored by the formation of strong intermolecular hydrogen bonds. Indeed, we have calculated favorable dimerization energies of $-16.0 \text{ kcal}\cdot\text{mol}^{-1}$ for Tt–Tt dimer and $-15.6 \text{ kcal}\cdot\text{mol}^{-1}$ for Tt–Ct dimer. Dimer adsorption on SiO_2 is favorable. We obtain a binding energy for the dimer of $-22.9 \text{ kcal}\cdot\text{mol}^{-1}$ for Tt–Tt and $-31 \text{ kcal}\cdot\text{mol}^{-1}$ for Tt–Ct (route 1 on Figure 2). In addition to the intermolecular hydrogen bond, two more hydrogen bonds are formed with the silanol groups on the surface (see Figure 7). Interestingly, the binding energy for the dimer is comparable to twice the binding energy of a monomer, which suggests that dimer adsorption is possible also at relatively low coverage. Experimentally, the presence of adsorbed dimers is confirmed by the appearance in the spectrum of a peak around 1709 cm^{-1} which corresponds to a very strongly red-shifted C=O acid,

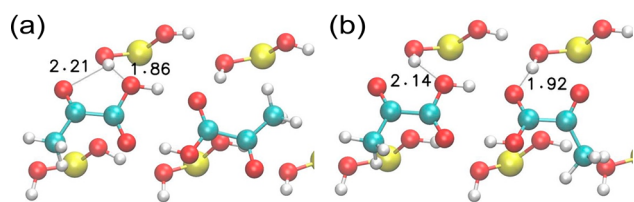


Figure 7. Top view of a snapshot extracted from AIMD simulations with the surface silanols and the dimer configurations: Tt–Tt (a) and Tt–Ct (b).

due to the strong hydrogen bond between two pyruvic acid molecules.

Humid Conditions: Experiment and Theory. To investigate the effect of surface adsorbed water ($\text{H}_2\text{O}(\text{a})$), the adsorption of pyruvic acid (100 mTorr) on silica was systematically studied at the following RHs: <1% (dry), 5.1%, 10.1%, 19.9%, 35.8%, 49.5%, and 81.0%. The amount of water adsorbed on the surface increases with RH (see Figure 8), as evident from

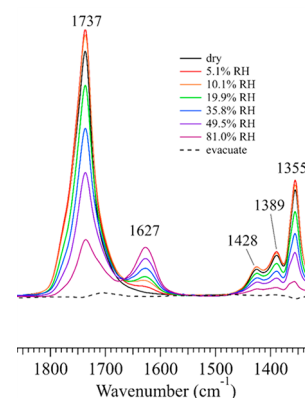


Figure 8. Absorbance spectra (ca. 1280 to 1860 cm^{-1}) of silica particles after first exposure to 100 mTorr pyruvic acid and then water vapor as a function of relative humidity (RH). The dashed line (— —) indicates the surface adsorbed species disappear after evacuation of both water and pyruvic acid.

the observed increase in intensity of the band at 1627 cm^{-1} (OH bending mode of water). The intensities of peaks attributed to adsorbed pyruvic acid (1737 , 1428 , 1389 , and 1355 cm^{-1}), however, decreased above 19.9% RH, suggesting that water competes for surface adsorption sites with pyruvic acid/displaced adsorbed pyruvic acid. This displacement role dominates at higher RH (>20%). An absence of pyruvate peaks around $1400\text{--}1600 \text{ cm}^{-1}$ (C=O stretching of pyruvate) was observed, indicating that adsorbed pyruvic acid did not deprotonate on the silica surface even in the presence of water. The disappearance of surface adsorbed species following evacuation suggests that the adsorption of pyruvic acid in the presence of water vapor is a reversible process as well.

We study the adsorption of Tc and Tt conformers under humid conditions using the same setup as in the previous section with an additional layer of water on the surface. The monomer adsorbs on the surface through a water bridge (see Figure 9). This water forms a strong HB to the hydrogen of the carboxyl group of the pyruvic acid (see Figure 10). Due to the presence of water at the interface, all the frequencies of the C=O stretches are red-shifted (see Table 2).

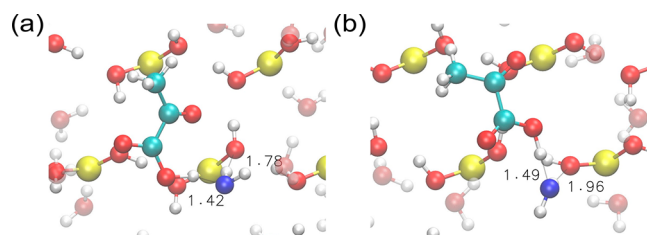


Figure 9. Snapshots of the top view of the Tc (a) and Tt (b) conformers under wet conditions. The water "bridge" is highlighted in blue.

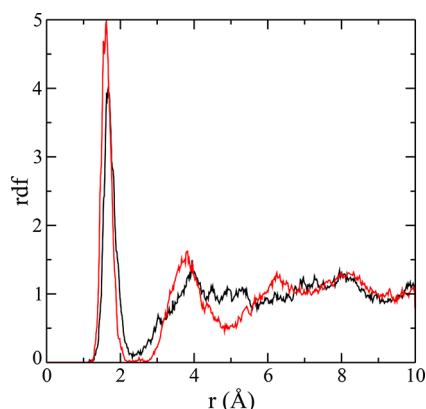


Figure 10. Radial distribution functions between the hydrogen atom of the pyruvic acid (black: Tc; red: Tt) and the oxygen atoms of the water solvent.

Curve Fitting Based on Theoretical Calculated Vibrational Frequencies. Dry Conditions. To further determine correct peak compositions and areas from experimental results, peak fitting was applied on the basis of the vibrational frequencies calculated in the theoretical section. In the gas phase, an overlap was observed for C=O stretching of the pyruvic acid ketone stretching at 1737 cm^{-1} and the Tt conformer (1760 cm^{-1}), respectively, as shown in Figure 4. When adsorbed on the surface, gas-phase Tc conformer acid stretching is also red-shifted from 1805 cm^{-1} to the shoulder at 1776 cm^{-1} , further overlapping with other C=O stretching species. By applying peak fitting to spectra ($1550\text{--}1900\text{ cm}^{-1}$) observed for silica particles after exposure to pyruvic acid ($\text{RH} < 1\%$), based on blueshift of the theoretical modeling, calculated vibrational frequencies were corrected by 60 cm^{-1} at 1756 , 1754 , 1736 , 1733 , and 1709 cm^{-1} , respectively. Note that the experimental measured gas-phase pyruvic acid vibrational frequencies are $60\text{--}65\text{ cm}^{-1}$ higher than calculated value. This systematic difference can possibly be attributed to the GGA functional used here, which may systematically soften the stretching modes. As shown in Figure 11, the adsorption band at 1756 cm^{-1} is assigned to C=O acid stretching of the adsorbed pyruvic acid Tc conformer, while 1754 cm^{-1} is due to C=O ketone stretching of adsorbed Tc conformer and pyruvic acid dimers (Tt–Ct). C=O acid stretching of the pyruvic acid Tc conformer has been shifted by 49 cm^{-1} from the gas phase as the acid COOH group is now involved in hydrogen bonding with hydroxyl groups on the silica surface. In contrast, Tt conformer C=O acid and ketone stretching shifted by 24 and 27 cm^{-1} , respectively, suggest that both Tt acid –COOH and ketone groups are involved in weak binding with surface hydroxyl groups. The broader peak at 1709 cm^{-1} is due to C=O acid stretching of hydrogen bonded pyruvic acid dimers. A previous laboratory study⁵⁴ has shown that other carboxylic acids, such as acetic acid, can form dimers on the silica surface. Pyruvic acid can form dimers,^{41,64} supporting our observation of the formation of a pyruvic acid dimer with a vibrational frequency at 1709 cm^{-1} . This observation can also be supported by theoretical modeling as discussed in the previous theoretical section.

The relative composition (%) (after peak fitting) of adsorbed pyruvic acid (Tt, Tc, and dimer) on the silica surface as a function of gas-phase pyruvic acid pressure is represented as the ratio of peak areas of dimer to monomers (sum of Tt and Tc conformers) in Figure 12. The C=O acid

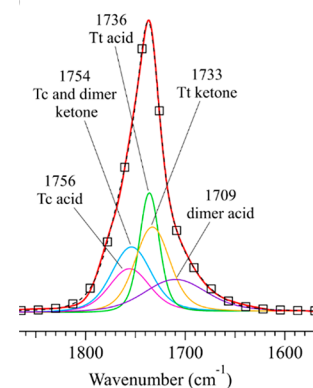


Figure 11. Absorbance spectra (from ca. 1560 to 1860 cm^{-1}) after peak fitting, for SiO_2 particles following exposure to 100 mTorr pyruvic acid ($\text{RH} < 1\%$). Absorbance bands correspond to C=O stretching of multiple species present on the surface, including: adsorbed pyruvic acid Tc conformer C=O acid stretch (1756 cm^{-1}); dimer and Tc conformer C=O ketone stretch (1754 cm^{-1}); Tt conformer C=O acid stretch (1736 cm^{-1}); Tt conformer C=O ketone stretch (1733 cm^{-1}); dimer acid C=O stretch (1709 cm^{-1}). The red line represents the original spectrum, and the dashed line with open square marker represents the overall fit.

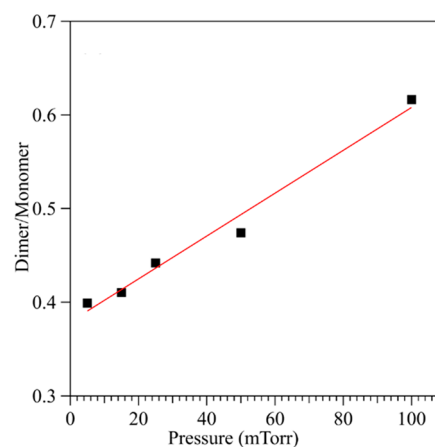


Figure 12. Ratio of peak area of C=O acid stretch from hydrogen bonded pyruvic acid dimers (1709 cm^{-1}) to total peak area of monomers (the sum of peak areas of the Tc acid (1756 cm^{-1}) + Tt acid (1736 cm^{-1})) as a function of gas-phase pyruvic acid pressure.

stretch is used to calculate the dimer/monomer ratio owing to the fact that the C=O ketone stretch of Tc and dimer overlap at 1754 cm^{-1} . The relative amount of pyruvic acid dimer formed on the silica surface increases with pressure (see Figure 12), while the total amount of pyruvic acid monomers decreases. Even at 1 Torr of pyruvic acid, no dimer is observed in the gas phase. Pyruvic acid is unlikely to form the dimer in the gas phase due to the intramolecular hydrogen bonding in the most stable Tc conformer. Figure 12 shows that, for molecularly adsorbed pyruvic acid on SiO_2 , the intensity of the infrared absorbance (represented by the peak area after peak fitting) due to dimers (at 1709 cm^{-1}) is comparable with the sum of monomers even for very low pyruvic acid pressure (5 mTorr). This suggests that SiO_2 favors the formation of dimers on the surface. Tang et al.⁵⁴ investigated heterogeneous reactions of acetic acid on SiO_2 surfaces and suggested that the acetic acid dimer is favorable in the presence of silica surfaces even under conditions when

monomers are the major species in the gas phase. This supports our observations for pyruvic acid (another carboxylic acid) adsorption on silica surfaces. As predicted by theoretical modeling, hydroxyl groups from the silica surface form more than one hydrogen bond with the pyruvic acid Tc conformer.

Humid Conditions. To further determine the experimental vibrational frequencies of the conformers of pyruvic acid in the presence of water, curve fitting has been applied on the basis of calculated vibrational frequencies (see Table 2). As shown in Figure 13, Tc conformer water bridged with the silica surface

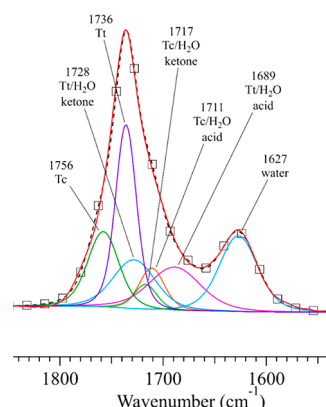


Figure 13. Absorbance spectra (from ca. 1550 to 1850 cm^{-1}) after peak fitting, for SiO_2 particles following exposure to 100 mTorr pyruvic acid (RH = 50%). Absorbance bands correspond to $\text{C}=\text{O}$ stretching of multiple species present on the surface, including: the adsorbed pyruvic acid Tc conformer (1756 cm^{-1}); Tt conformer (1736 cm^{-1}); water bridged Tt conformer ketone stretch (1728 cm^{-1}); water bridged Tc conformer ketone stretch (1717 cm^{-1}); water bridged Tc conformer acid stretch (1711 cm^{-1}); water bridged Tt conformer acid stretch (1689 cm^{-1}). The absorbance band centered at 1627 cm^{-1} is assigned to the OH bending mode of water. The red line represents the original spectrum, and the dashed line with open square marker represents the overall fit.

has been observed at 1711 and 1717 cm^{-1} for acid and ketone stretches, respectively, while Tt conformer water bridged with the silica surface results in two $\text{C}=\text{O}$ stretches at 1728 cm^{-1} (ketone) and 1689 cm^{-1} (acid). The peak at 1627 cm^{-1} can be assigned to the OH bending mode of water. Note that water vapor is introduced following pyruvic acid adsorption and reaches equilibrium on the silica surface. Therefore, the adsorbed Tt conformer (1736 cm^{-1}) and Tc conformer (1756 cm^{-1}) still exist on the surface. Note that acid and ketone stretches are not distinguished here as they are close in frequency when Tt and Tc conformers are directly interacting with the silica surface. Water-bridged pyruvic acid Tc conformer red-shifted by 45 cm^{-1} (acid) and 37 cm^{-1} (ketone) suggests that the Tc conformer acid and ketone $\text{C}=\text{O}$ are both involved in hydrogen bonding with adsorbed water. On the other hand, the Tt pyruvic acid conformer interacts with water by its acid $\text{C}=\text{O}$ group, suggested by the 47 cm^{-1} redshift of the acid group and the 5 cm^{-1} shift of the ketone stretch. The redshift of Tc and Tt conformers suggests that pyruvic acid forms a significantly stronger hydrogen bond with adsorbed water.

CONCLUSIONS

Transmission FTIR spectroscopy and theoretical simulations were used to investigate the heterogeneous interactions of

pyruvic acid with hydroxylated SiO_2 surfaces. To investigate the effect of SiO_2 on the configuration of pyruvic acid, experiments were carried out by exposing the silica surface to pyruvic acid as a function of pressure. Pyruvic acid adsorption on the silica surface is a molecularly reversible process. Pyruvic acid has two stable configurations in the gas phase: Tc and Tt conformers. Following adsorption, both conformers (Tc and Tt) exist on the surface forming two hydrogen bonds. The Tc conformer binds through $\text{C}=\text{O}$ and OH (from $-\text{COOH}$) with silica surface hydroxyl groups, while the Tt conformer binds through both $\text{C}=\text{O}$ acid and $\text{C}=\text{O}$ ketone groups. Even at very low pyruvic acid pressure where no pyruvic acid dimer has been observed in the gas phase, pyruvic acid dimer is observed on the surface following adsorption. Tt–Tt and Tt–Ct are stable adsorbed pyruvic acid dimers on the silica surface. Interestingly the binding energy of a dimer on the surface is comparable to twice the monomer binding energy, which suggests that dimers may be found on the surface even at relatively low concentrations. The role of water was studied as a function of relative humidity. Water is found to compete for surface adsorption sites with pyruvic acid at all RHs studied. In addition, calculations reveal that adsorbed pyruvic acid Tt and Tc conformers bind with silica surface hydroxyl groups through a water bridge in the presence of adsorbed water.

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Notes

The authors declare no competing financial interest.

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